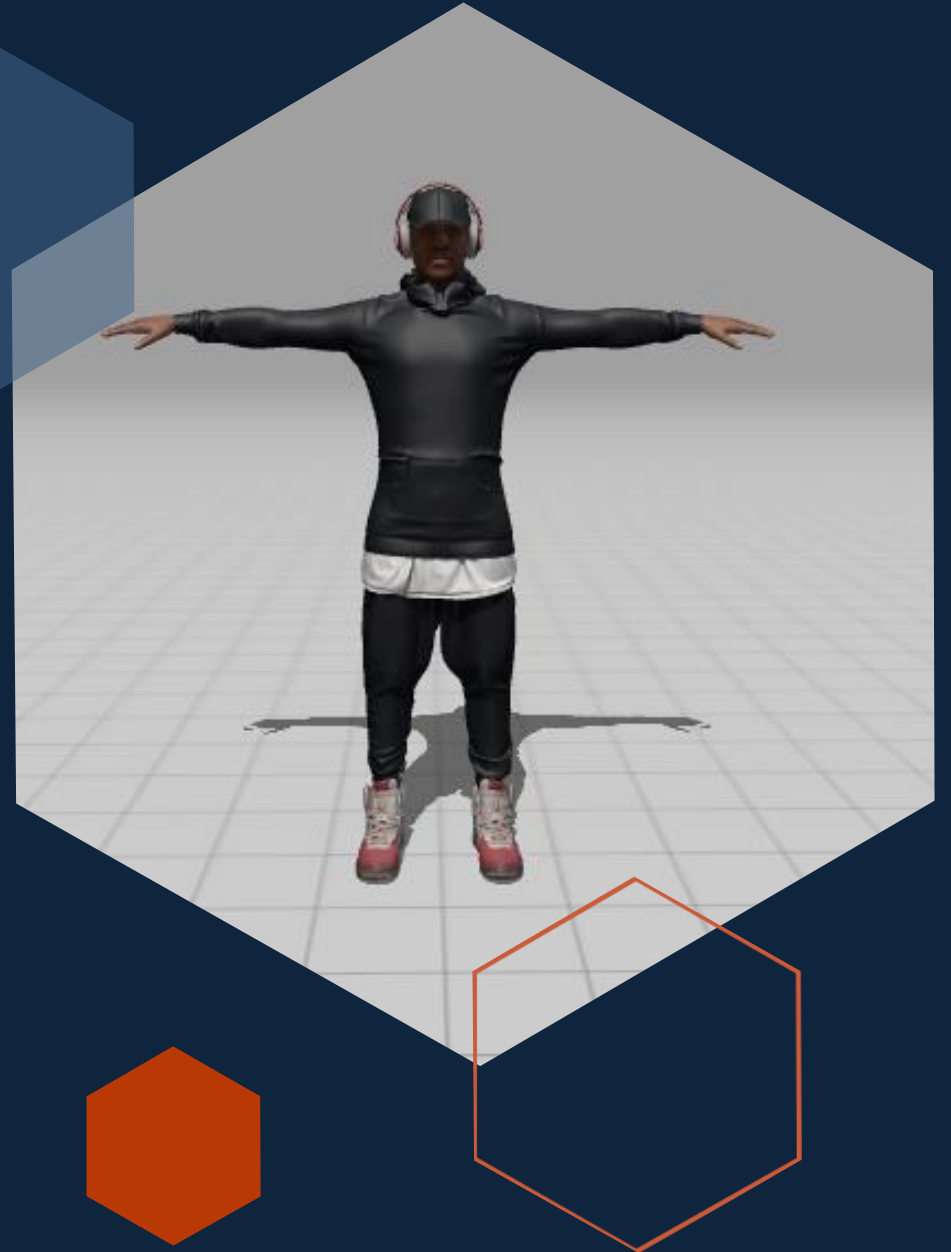


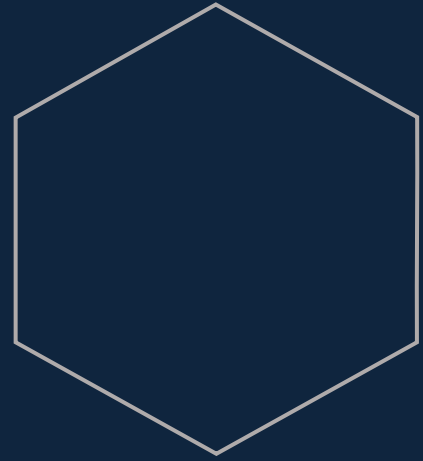
Presentation title : Vector

Presenter Names : Vishal Kasapurambodagala

Anudeep Puram

Vikas Nayikoti





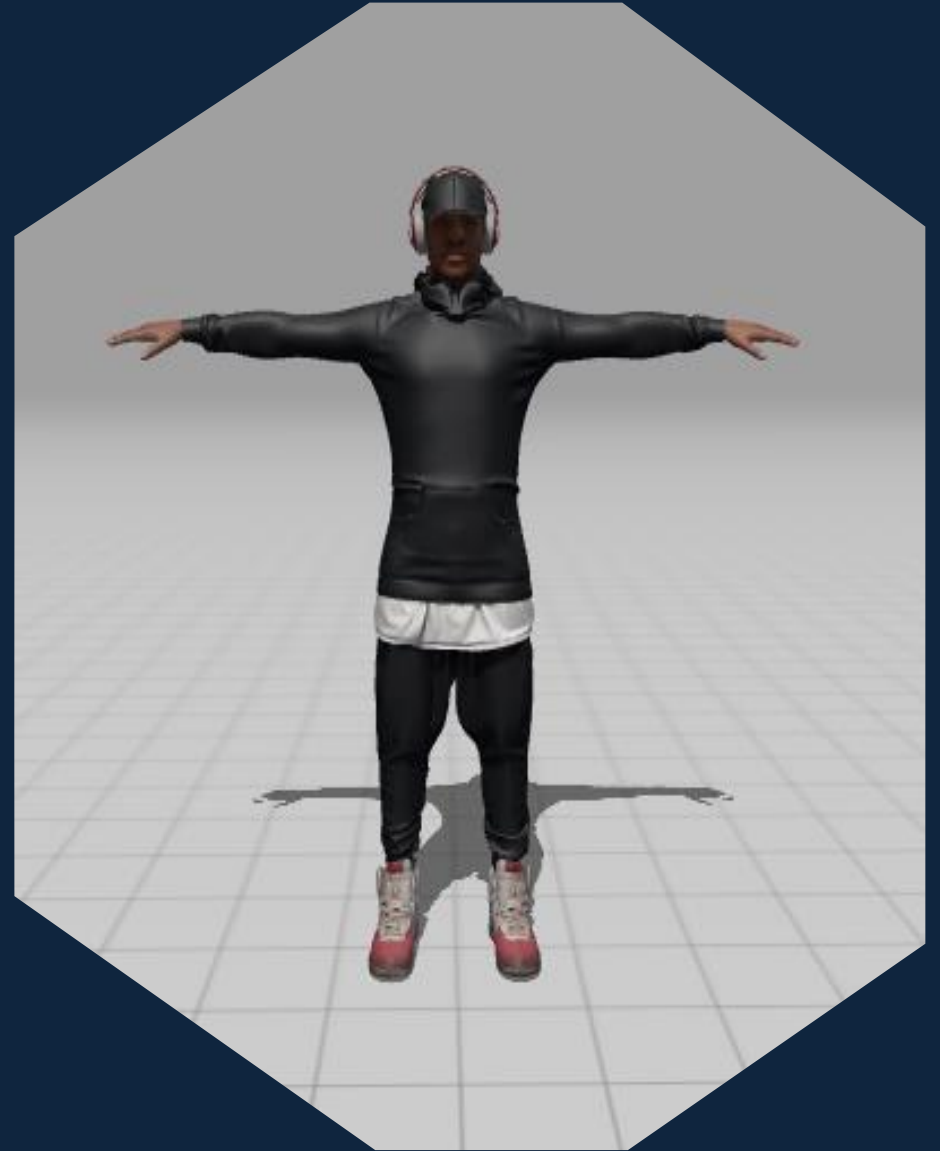
Agenda





Introduction

1. We build a character named Ramu.
2. This is a project where similarly work as subway surfers.
3. We developed whole project in Unreal Engine.

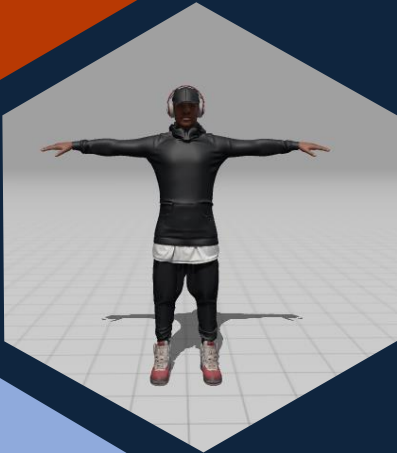


Primary goals

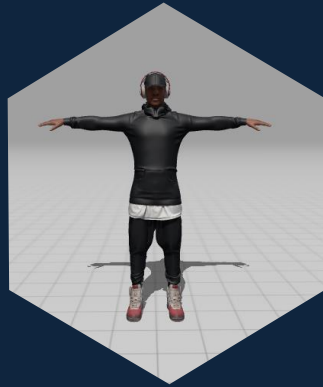
Character Rigging

Adding Animation to the character

Reaction of character
with the surroundings

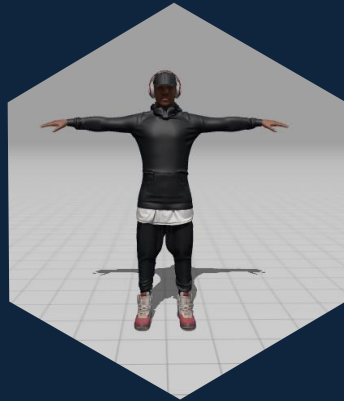


Primary goals



Character Rigging

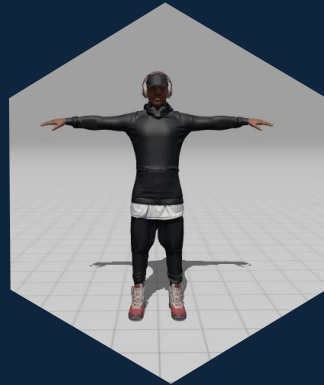
- ❑ We used Character Rigging, which it is like giving life to a character in animations.
- ❑ It is the process of setting up the character's "Skeleton" so it can move in a smooth and realistic way.
- ❑ This allows us to control how a character's different parts, like arms and legs move and interact.



Adding animation to the character

- ❑ Adding animations to a character involves creating a series of movement that bring the character to life.
- ❑ **Walk :** We rig the character to move it's leg in a walking motion then we loop this animation seamlessly, which creates an illusion of walking but we make sure character is smooth and natural.

Primary goals



Adding animation to the character

- ❑ **Run** : Run is similar to walk, but involves in exaggerating the leg movements in a faster pace.

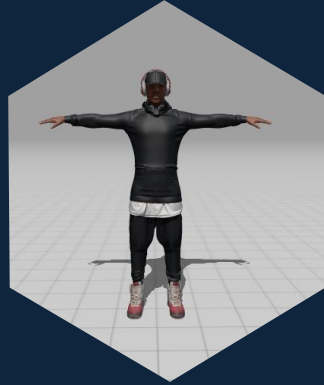
The posture of the character may lean forward slightly to convey the speed and momentum.

- ❑ **Jump** : Action of jumping involves in the character bending in air and moving to it's current state.

We need to bend the knees of the character pushing off the ground and lifting into the air.

Mid-air animations may include movements of the arms and legs to convey natural jumping motion.

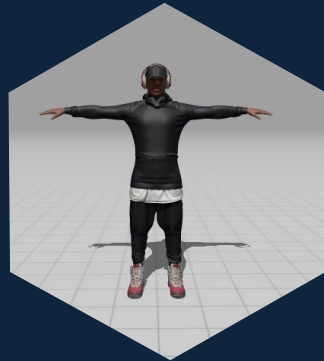
Primary goals



Adding animation to the character

- ❑ **Hit & Fall** : When a character is involved to hit an obstacle in map which gives an impact which involve character recoiling.

The character to loose the balance and fall to the ground then that would convey a scene of uncontrolled descent.



Reaction of character with the surroundings

- ❑ The animation of character interacts with the surroundings based on the objects and obstacles present in the map.
- ❑ This involves in creating movements and behavior accordingly that respond to the environment.
- ❑ This includes responses like looking around, reaching for objects to match the environment, making the character feel more alive.

Technical Results

Efficient Rigging

Performance Optimization

Realism

Responsive Interaction

Consistent Artistic Style



Results

Efficient Rigging	Responsive Interaction	Performance optimization
• Rigging process is an technical aspect of character animation, is well-constructed and adaptable. • This allows us to easily manipulate and control movements resources during the animation process.	• The character is responding dynamically to the given input or changes made in the environment. • Resulting to adapt different scenarios, which gives us more immersive experience. • Based on responsive interaction, the performance is rendered accordingly.	• Efficient rigging and animation gives the best optimization of the character. • The optimized animations enhance the overall performance in rendering and playback.

Results

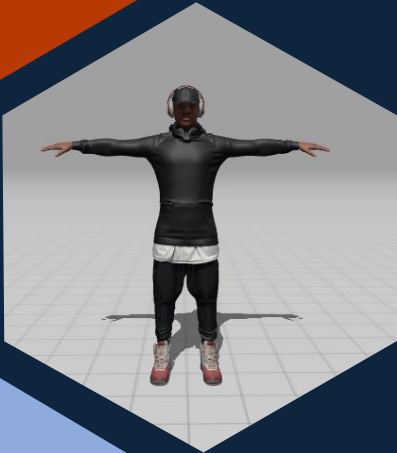
Consistent Artistic Style	Realism
• The style of execution to the character aligned perfectly as per the given input. • This involves consistent proportions, styles and movements that ensures to more artistic experience.	• The technical aspects such as Walk, Run, Jump , Hit & Fall that involved in smooth transition. • This aspect of contribution resembles the overall experience in successful character animation Realistically.

Conclusion

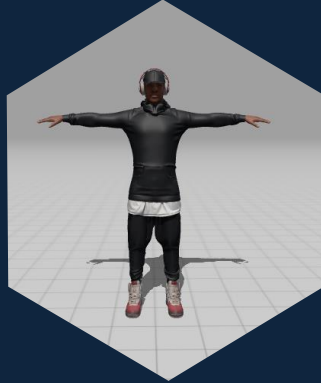
Animation Principles

Keyframes

Inverse Kinematics

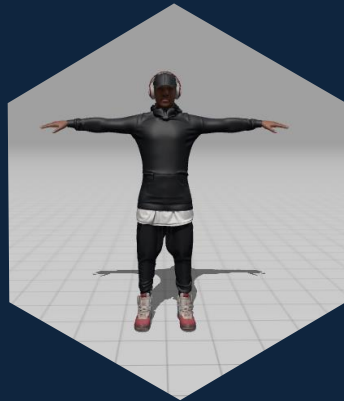


Animation Principles



- ❑ The basic principle of the animation are designed to be seamless with smooth transitions between the different actions to give us natural feel.
- ❑ The timing and space we use gives us more realistic motion.

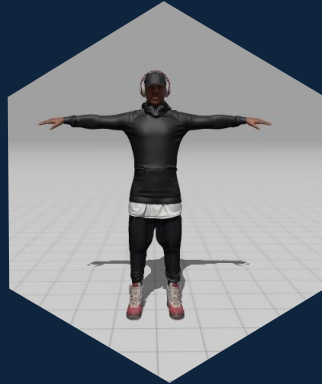
Conclusion



Key Frames

- ❑ Key frames are more important gives us the smooth motion when matching all the other sequence match frames.
- ❑ Key frames establish the positions of a character at specific moments in time.
- ❑ They can be various set of properties, scale, rotation and opacity.

Conclusion



Inverse Kinematics

- ❑ This inverse kinematics is a technique helped in rigging that establishes a relationship between the movement of one part of a character body and subsequent of other parts.
- ❑ In similar terms, a character's hand can automatically adjust the position of the the forearm and upper arm.

"Thank You"

Summary

At Contoso, we believe in giving 110%. By using our next-generation data architecture, we help organizations virtually manage agile workflows. We thrive because of our market knowledge and great team behind our product. As our CEO says, “Efficiencies will come from proactively transforming how we do business.”

